Opgaver kapitel 12

12.31

25. Figure 12.17*a* shows an outstretched arm with mass 4.3 kg. The arm is 56 cm long, and its center of gravity is 21 cm from the shoulder. The hand at the end of the arm holds a 6.0-kg mass. (a) Find the torque about the shoulder due to the weight of the arm and the 6.0-kg mass. (b) If the arm is held in equilibrium by the deltoid muscle, whose force on the arm acts below the horizontal at a point 18 cm from the shoulder joint (Fig. 12.17*b*), what's the force exerted by the muscle?

BIO

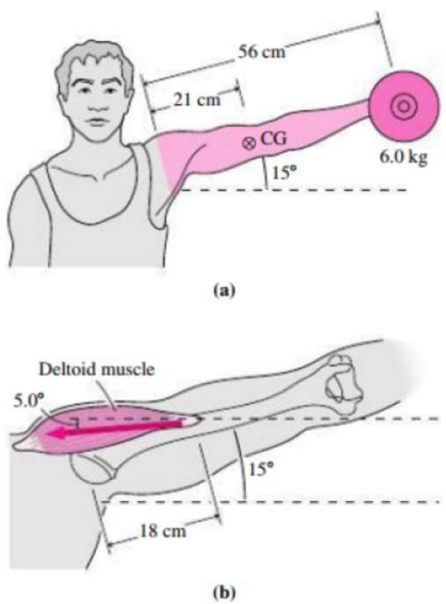

FIGURE 12.17 Problem 25

12.32

26. A uniform sphere of radius R is supported by a rope attached to a vertical wall, as shown in Fig. 12.18. The rope joins the sphere at a point where a continuation of the rope would intersect a horizontal line through the sphere's center a distance $\frac{1}{2}R$ beyond the center, as shown. What's the smallest possible value for the coefficient of friction between wall and sphere?

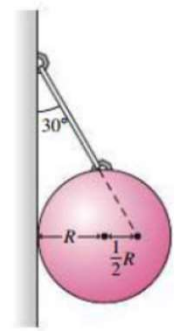

FIGURE 12.18 Problem 26

12.36

30. The boom in the crane of Fig. 12.21 is free to pivot about point P and is supported by the cable attached halfway along its 18-m length. The cable passes over a pulley and is anchored at the back of the crane. The boom has mass 1750 kg distributed uniformly along its length, and the mass hanging from the boom is 2200 kg. The boom makes a 50° angle with the horizontal. Find the tension in the cable.

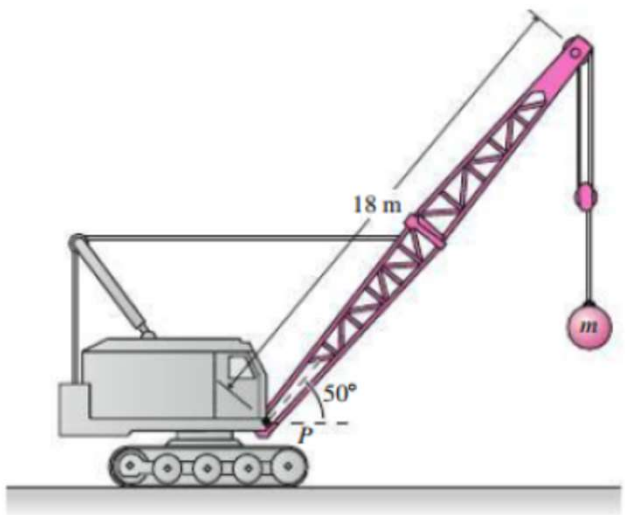

FIGURE 12.21 Problem 30

12.38

38. Figure 12.24 shows a 1270-kg car that has slipped over an embankment. People are trying to hold the car in place by pulling on a horizontal rope. The car's bottom is pivoted on the edge of the embankment, and its center of mass lies farther back, as shown. If the car makes a 34° angle with the horizontal, what force must the people apply to hold it in place?

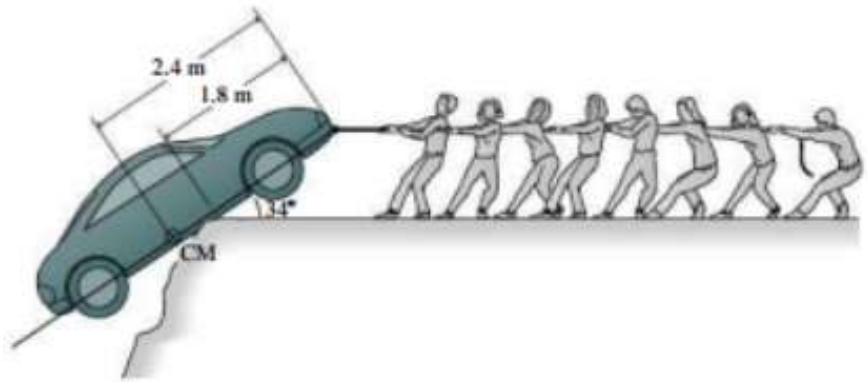

FIGURE 12.24 Problem 38

12.67

63. DATA Engineers designing a new semiconductor device measure the potential energy that results when they move an electron to different positions within their device. The device is one-dimensional, so the positions all lie along a line. The table below gives the resulting data. Plot these data and from your plot, determine the approximate positions of any equilibria and whether such equilibria are stable or unstable.

Position x (nm)	0	3.26	5.85	6.41	7.12	9.37	10.5	12.2	14.0	14.5	15.3	17.2
Potential energy U (aJ)	1.5	0.65	0.30	0.47	0.85	2.7	3.3	2.1	−0.47	−0.86	−0.72	3.2